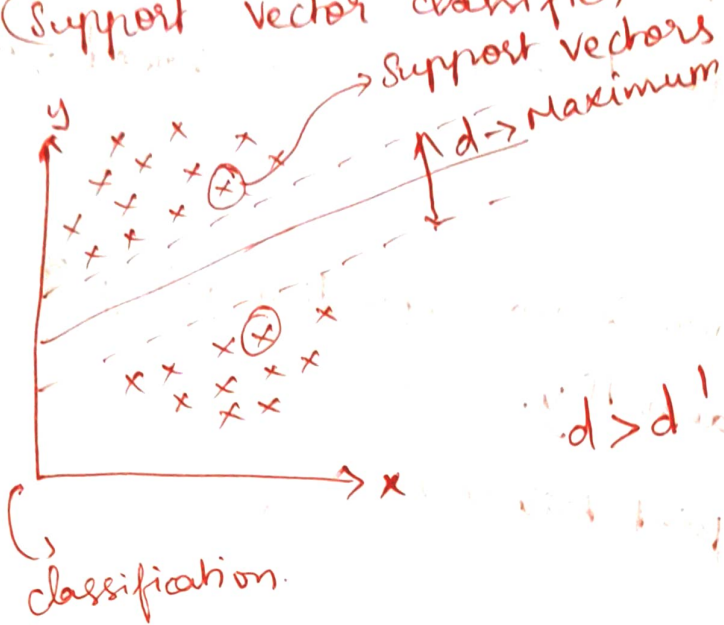


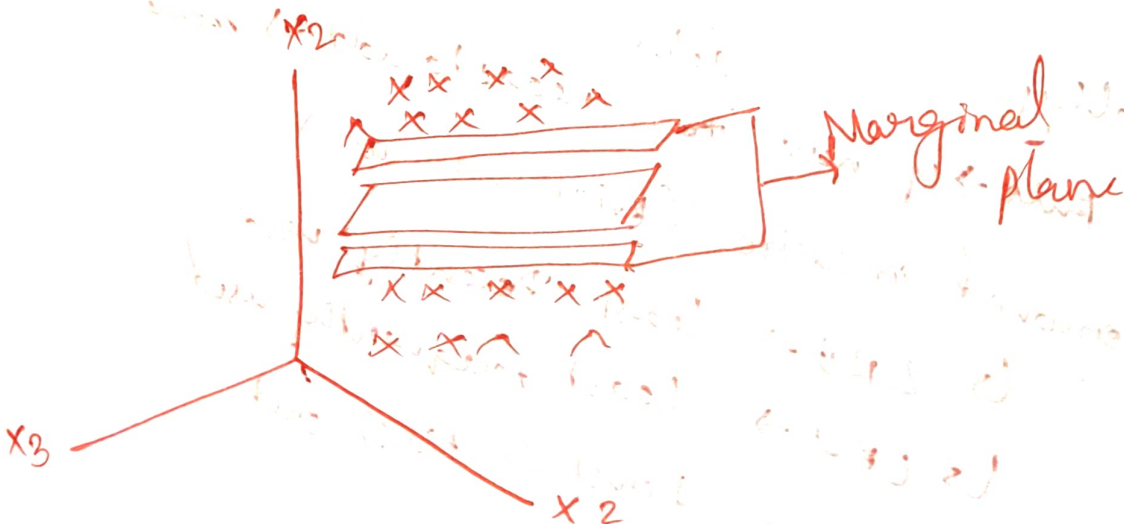
Machine learning

Support Vector Machines.

SVC (Support vector classifier)



Aim: To create a Marginal plane
and Support vector points
to categorize / clearly
visible of for the new points



If linear use linear kernel
if non linear use rbf kernel } best kernel

Soft Margin and Hard Margin

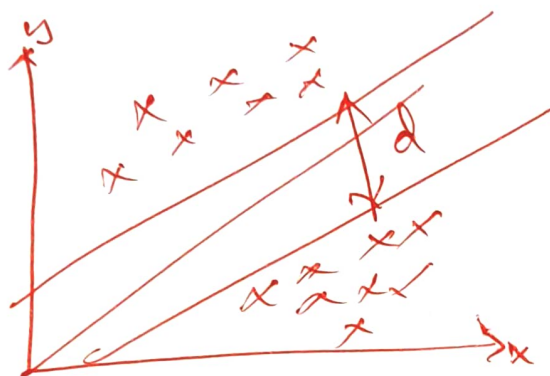
In Real World problem



→ Soft Margin

we have overlapping of points
so error will be huge.

Hard Margin



The distance is huge so
error will be less.

-ve $d_i = -ve$ below the plane
 $d = +ve$ above the plane

maximize

w, b

changing

$\frac{2}{\|w\|}$

→ magnitude.

$$y_i \begin{cases} +1 & w^T x + b \geq 1 \\ -1 & w^T x + b \leq -1 \end{cases}$$

constant → $y_i * (w^T x + b) \geq 1$

$\sum_i$ → Summation of the distance of
the incorrect data points from
the Marginal plane